

JEE MAIN REVISION MASTER: HEIGHT AND DISTANCE OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 40 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Height And Distance — Standard Calculations & Problem Archetypes	23 PYQs	Quantitative Analysis
High	Height And Distance — Core Principles & Governing Laws	10 PYQs	Core Principle
High	Height And Distance — Applied Cases, Reactions & Traps	8 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Height And Distance — Standard Calculations & Problem Archetypes	$AB=x$, $\quad \frac{80}{x}$, $\quad \tan 45^\circ = \frac{PT}{ST} = \frac{5}{ST}$
Height And Distance — Core Principles & Governing Laws	$\tan 45^\circ = \frac{PT}{ST} = \frac{5}{ST}$, $\quad NA = \sqrt{15^2 - 9^2} = 12$, $\quad = 5(2 + \sqrt{3})$
Height And Distance — Applied Cases, Reactions & Traps	$\frac{\square}{BM} = \frac{1}{\sqrt{7}}$, $\quad \frac{\square}{AM} = \frac{1}{3\sqrt{2}}$, $\quad \rightarrow t = \frac{2\sqrt{2t(1-t^2)}}{1+\sqrt{7}} \cdot \frac{2t}{1-t^2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Height And Distance — Standard Calculations & Problem Archetypes (Very High Yield — 23 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Height And Distance.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due North of the tower is 30° . Let the height of the tower be h m. Then the distance between the two persons is $\frac{h}{\sqrt{3}}$ m." [Ans: N/A]
- [2022 Evening Shift] Q#1 (MCQ): "The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45° . Let R be a point on the tower such that $AR = 10$. Then the angle of elevation of P from R is α ." [Ans: N/A]
- [2022 Morning Shift] Q#1 (MCQ): "Let AB and PQ be two vertical poles, 160 m apart from each other. Let C be the middle point of B and Q , which are feet of these two poles. $AC = 100$ m. Then the height of the pole AB is h m." [Ans: N/A]

3.2 Height And Distance — Core Principles & Governing Laws (High Yield — 10 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Height And Distance.

■ ■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Height And Distance.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Height And Distance PYQs.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due North of the tower is 30° . Let the height of the tower be h m. Then the distance between the two persons is $\frac{h}{\sqrt{3}}$ m." [Ans: N/A]
- [2022 Morning Shift] Q#3 (MCQ): "The angle of elevation of the top of a tower from a point A due north of it is α and from a point B at a distance of 9 units due west of A is β . Then the height of the tower is $\frac{9 \sin \alpha \sin \beta}{\cos \alpha - \sin \beta}$ units." [Ans: N/A]
- [2022 Morning Shift] Q#3 (MCQ): "A tower PQ stands on a horizontal ground with base Q on the ground. The point R divides the tower in two parts such that $QR = 15$ m. If the angle of elevation of P from R is 45° and the angle of elevation of P from Q is 60° , then the height of the tower is h m." [Ans: N/A]

3.3 Height And Distance — Applied Cases, Reactions & Traps (High Yield — 8 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Height And Distance.

■ ■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Morning Shift] Q#2 (MCQ): "From the top A of a vertical wall AB of height 30 m, the angles of depression of the top P and bottom Q of a vertical tower PQ are 30° and 60° respectively. Then the height of the tower is h m." [Ans: N/A]
- [2022 Morning Shift] Q#2 (MCQ): "Let a vertical tower AB of height 20 m stands on a horizontal ground. Let from a point P on the ground a man can see upto height 10 m of the tower. Then the distance of P from the base of the tower is h m." [Ans: N/A]
- [2021 Evening Shift] Q#2 (MCQ): "Let in a right angled triangle, the smallest angle be θ . If a triangle formed by taking the reciprocal of its sides is also a right angled triangle, then θ is $\frac{\pi}{6}$." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 11th April Evening Shift (Q#1)

The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due west of the tower is 30° . If the height of the tower is 5 meters, then the distance (in meters) between the two persons is equal to

- (A) 10
- (B) $(5/2)\sqrt{5}$
- (C) $5\sqrt{5}$
- (D) 5

Correct Answer: None

Solution Analysis: Let's denote the person standing due south as S and the one standing due west as W. Also, let the tower be at point T. From person S's perspective, we have a right triangle $\triangle SPT$. The height of the tower PT is given as 5m, which is the opposite side for angle S. The angle at S is 45° . From the tangent trigonometric ratio, we have : $\tan 45^\circ = (PT / ST) = (5 / ST)$ Since $\tan 45^\circ = 1$, we get $ST = 5$ m. Similarly, from person W's perspective, we have a right triangle $\triangle WPT$. The angle at W is 30° . From the tangent trigonometric ratio, we have: $\tan 30^\circ =$

Benchmark Problem 2: JEE Main 2023 (Online) 6th April Morning Shift (Q#2)

From the top A of a vertical wall AB of height 30m, the angles of depression of the top P and bottom Q of a vertical tower PQ are 15° and 60° respectively, B and Q are on the same horizontal level. If C is a point on AB such that $CB=PQ$, then the area (in m^2) of the quadrilateral BCPQ is equal to :

- (A) $200(3-\sqrt{3})$
- (B) $300(\sqrt{3}-1)$
- (C) $300(\sqrt{3}+1)$
- (D) $600(\sqrt{3}-1)$

Correct Answer: None

Solution Analysis: Given, AB be a vertical wall of height 30m and PQ be a vertical tower. Such that $\angle BQA = \angle TAP = 60^\circ$ (Alternate angles) and $\angle CPA = \angle TAP = 15^\circ$ Now, $\tan 60^\circ = (AB / BQ) = (30 / BQ) \Rightarrow \sqrt{3} = (30 / BQ) \Rightarrow BQ = \frac{30}{\sqrt{3}} = 10\sqrt{3}$ $\begin{aligned} &\& \tan 15^\circ = \frac{AC}{PC} \end{aligned}$ $\& \& \Rightarrow 2 - \sqrt{3} = \frac{AC}{10\sqrt{3}}$ (because $PC=BQ$) $\& \& \Rightarrow AC = 10\sqrt{3}(2 - \sqrt{3}) = 20\sqrt{3} - 30$ $\& \& \therefore BC = AB - AC = 30 - (20\sqrt{3} - 30) = 60 - 20\sqrt{3}$ $\& \& \end{aligned}$

Benchmark Problem 3: JEE Main 2022 (Online) 27th July Evening Shift (Q#1)

The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45° . Let R be a point on AQ and from a point B, vertically above R, the angle of elevation of P is 60° . If $\angle BAQ = 30^\circ$, $AB = d$ and the area of the trapezium PQRB is α , then the ordered pair (d, α) is :

- (A) $(10(\sqrt{3}-1), 25)$
- (B) $(10(\sqrt{3}-1), (25/2))$
- (C) $(10(\sqrt{3}+1), 25)$
- (D) $(10(\sqrt{3}+1), (25/2))$

Correct Answer: None

Solution Analysis: Let $BR = x$ ($x/d = 1/2$) $\Rightarrow x = (d/2)$ $\frac{10-x}{10-x\sqrt{3}} = \sqrt{3} \Rightarrow 10-x = 10\sqrt{3}-3x$ $2x = 10(\sqrt{3}-1)$
 $x = 5(\sqrt{3}-1)$ $d = 2x = 10(\sqrt{3}-1)$ $\alpha = (1/2)(x+10)(10-x\sqrt{3}) = \text{Area (PQRB)} = (1/2)(5\sqrt{3}-5+10)(10-5\sqrt{3}(\sqrt{3}-1)) = (1/2)(5\sqrt{3}+5)(10-15+5\sqrt{3}) - (1/2)(75-25) = 25$